ENME601 Thermodynamics

Week 6 – The Second Law of Thermodynamics

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Objectives

- Introduce the **second law of thermodynamics**.
- Discuss **thermal energy reservoirs, heat engines, refrigerators, and heat pumps** and their thermal efficiencies and coefficients of performance.
- Describe the **Kelvin–Planck** and **Clausius statements** of the second law of thermodynamics.
- Discuss **reversible and irreversible processes.**
- Describe the **Carnot cycle** and examine the **Carnot principles**, idealized Carnot heat engines, refrigerators, and heat pumps.
- Determine the expressions for the **thermal efficiencies** and **coefficients of performance** for **reversible heat engines, heat pumps, and refrigerators**.

Introduction

- The first Law requires that **energy be conserved during a process**. The Second Law of Thermodynamics goes further and asserts that **processes involving energy changes *occur in a certain direction*** and that **energy has *quality* as well as quantity**.
- A process cannot take place unless it **satisfies both the first law and the second law of thermodynamics**.
- These processes are okay according to the first law but violate the second law.

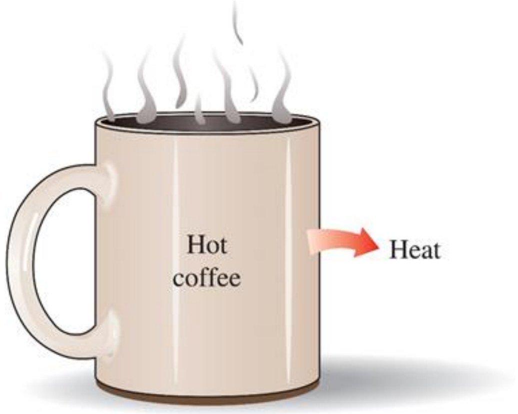


Figure 7–1: A cup of hot coffee does not get hotter in a cooler room.

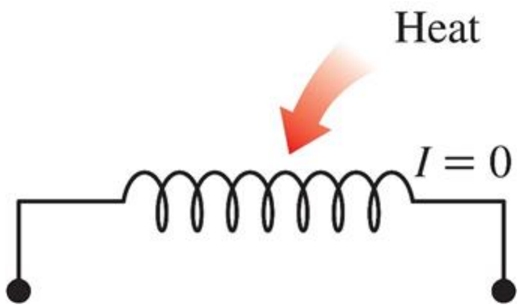


Figure 7–2: Transferring heat to a wire will not generate electricity.

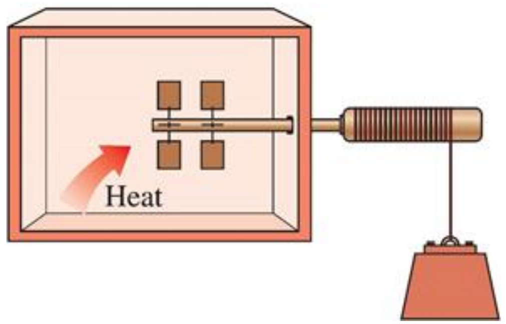
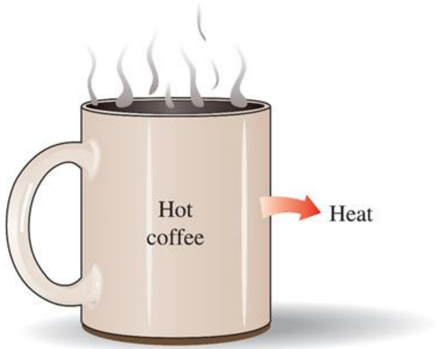


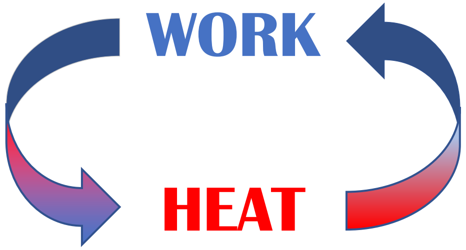
Figure 7–3: Transferring heat to a paddle wheel will not cause it to rotate.

Introduction

Major uses of the second law.

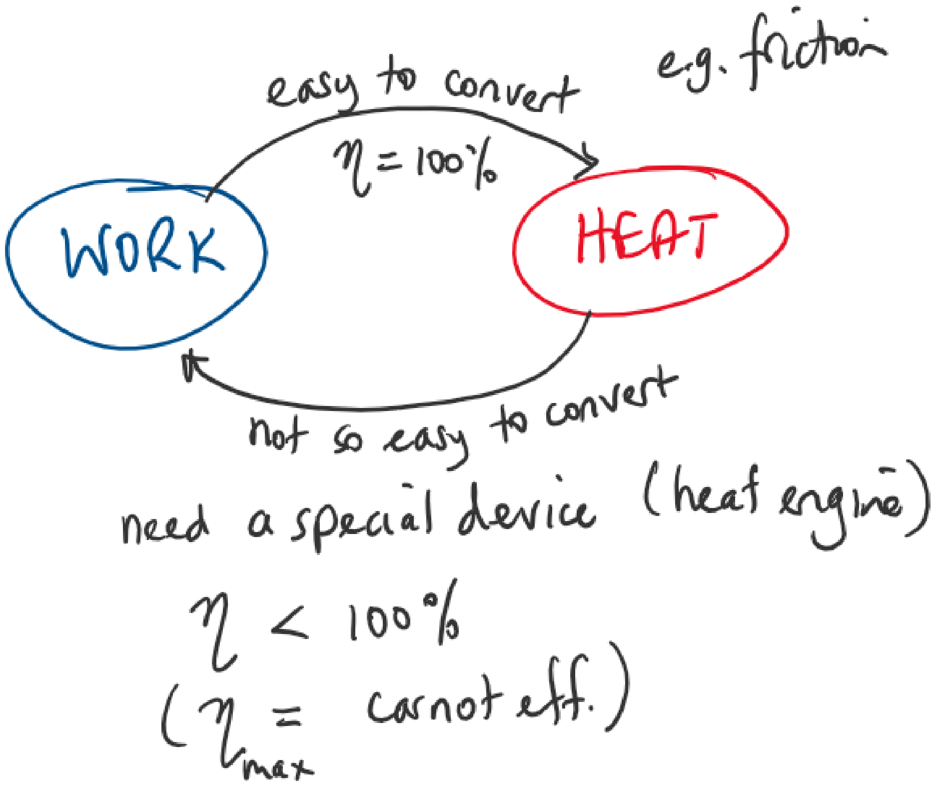
- The second law may be used to **identify the direction of processes**.
- The second law asserts **that energy has *quality* as well as quantity**. For example, we will find that work is a better form of energy flow than heat.
- The second law of thermodynamics is also used in determining the ***theoretical limits* for the performance** of commonly used engineering systems such as heat engines and refrigerators
- The second law can also predict the ***degree of completion* of chemical reactions**.

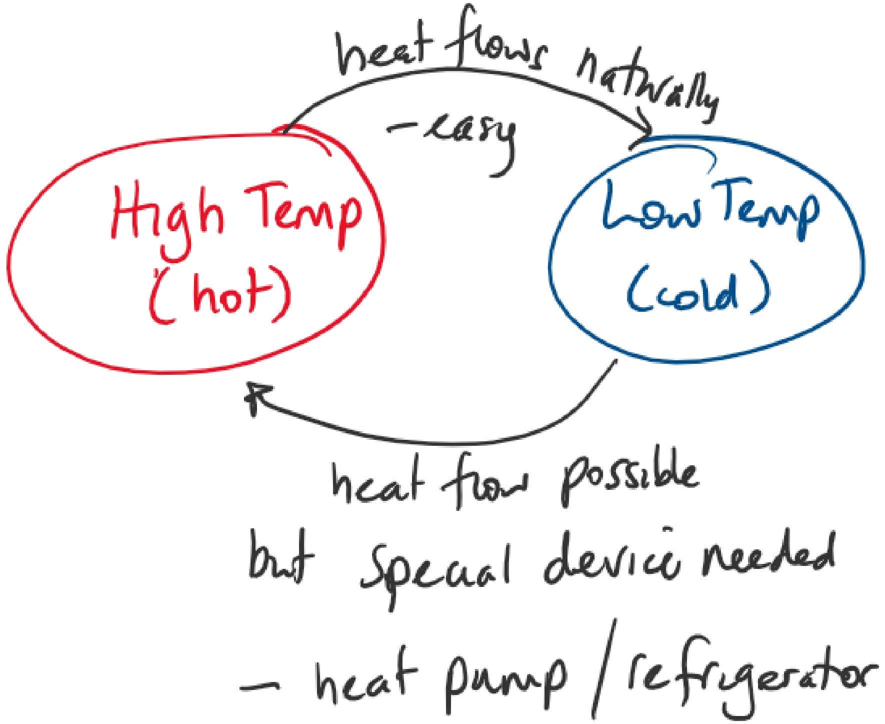




Introduction

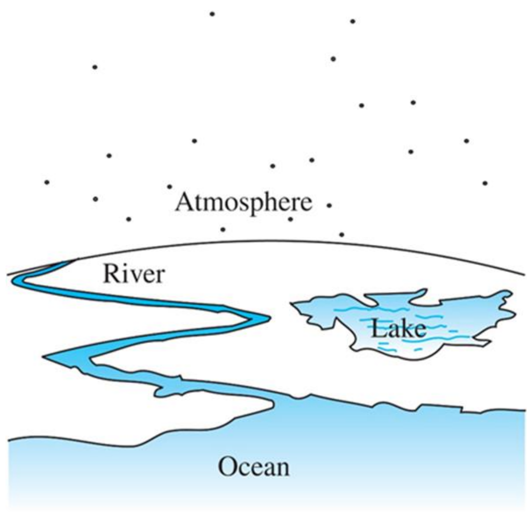
Illustrations of the second law.

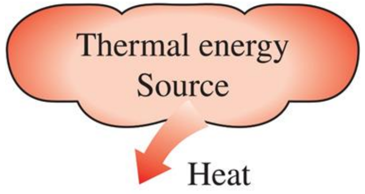


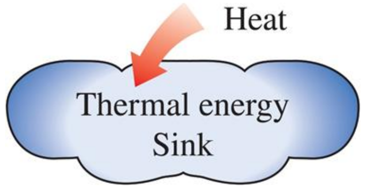


Thermal Energy Reservoirs (Sources and Sinks)

- A ***thermal energy reservoir*** (or just ***reservoir***) is a hypothetical body that has a **large thermal energy capacity** (mass × specific heat)
- It can **supply or absorb finite amounts of heat** energy without undergoing any appreciable temperature change.
- Examples: the atmosphere, lakes, rivers, the ocean
- A reservoir that **supplies heat energy is called a source**
- A reservoir that **absorbs heat energy is called a sink**

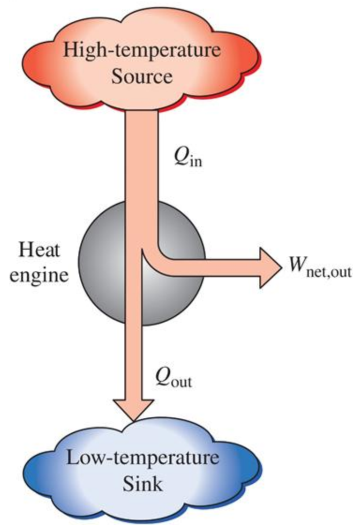


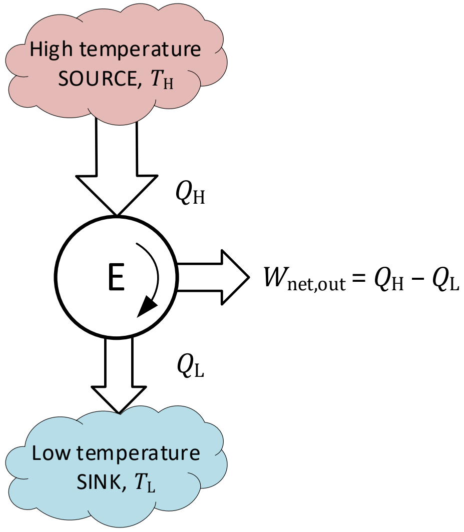




Heat Engines

- Work can be converted directly and completely to heat (friction) but **converting heat to work requires a special device called a *heat engine***.
- Heat engines are characterised by
  - They receive heat, Q_{H} from a high-temperature source (at T_{H}). (combustion of fuel, solar energy, nuclear reactor etc).
  - They convert part of this heat to work, $W_{\mathrm{net,out}}$ (e.g. rotating shaft).
  - They reject remaining waste heat, Q_{L} to a low-temperature sink(at T_{L}). (Usually the atmosphere, lake etc).
  - They operate on a cycle.
- Heat engines and other cyclic devices usually involve a fluid called the ***working fluid***. Heat is transferred to and from the working fluid as it undergoes the cycle.
- Note: $Q_{\mathrm{H}} = Q_{\mathrm{in}}$, $Q_{\mathrm{L}} = Q_{\mathrm{out}}$ (and are all magnitudes).

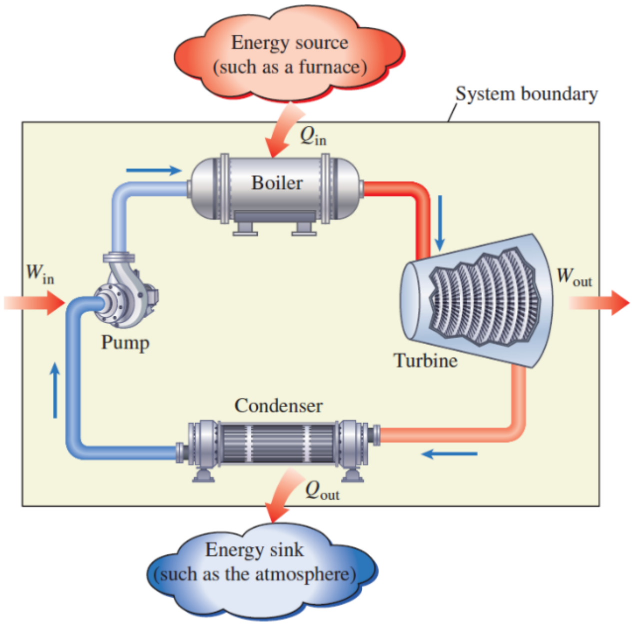


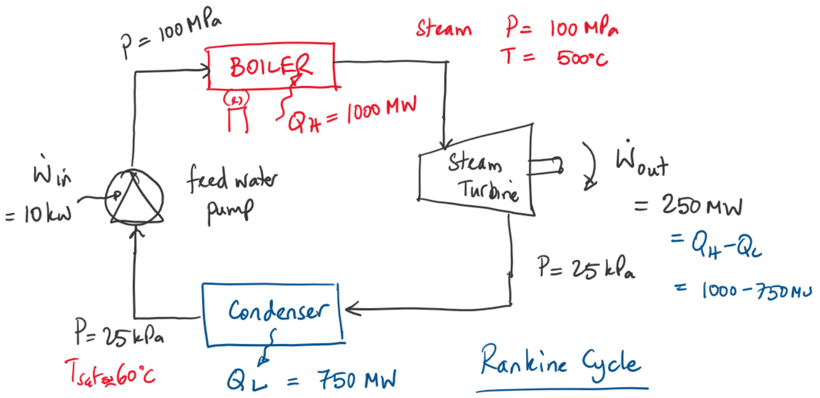


Heat Engines

Example: Steam Power Plant (Rankine Cycle)

- Heat is supplied to the cycle ($\dot{Q}_{\mathrm{H}}$) via the boiler (e.g. combustion of gas in the furnace). This increases the temperature and energy of the steam (e.g. 100 MPa, 500°C)
- The high energy steam enters the turbine which converts some of that energy into shaft work ($\dot{W}_{\mathrm{out}}$) and leaves at low pressure
- The steam condenses to liquid water in the condenser by rejecting heat ($\dot{Q}_{\mathrm{L}}$) to the environment.
- The liquid water at low pressure from the condenser enters the feed water pump which increases the pressure (compresses) the liquid water to boiler pressure ($\dot{W}_{\mathrm{in}}$).





$$W_{\mathrm{net,out}} = W_{\mathrm{out}} - W_{\mathrm{in}}$$

Heat Engines

Thermal Efficiency

- A measure of the **performance** of a heat engine
- Generally we can define performance as:

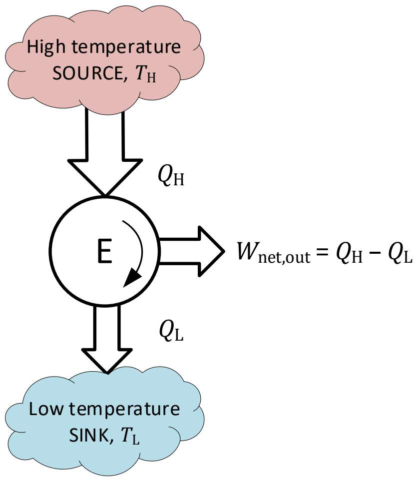
$$\text{performance} = \frac{\text{desired output}}{\text{required input}}$$

- For a heat engine the desired output is $W_{\text{net,out}}$ and the required input is Q_{H}. So the thermal efficiency of a heat engine, η_{th} is:

$$\boxed{\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{\text{H}}}}$$

- And from the first law: $W_{\text{net,out}} = Q_{\text{H}} - Q_{\text{L}}$, so

$$\boxed{\eta_{\text{th}} = \frac{Q_{\text{H}} - Q_{\text{L}}}{Q_{\text{H}}} = 1 - \frac{Q_{\text{L}}}{Q_{\text{H}}}}$$



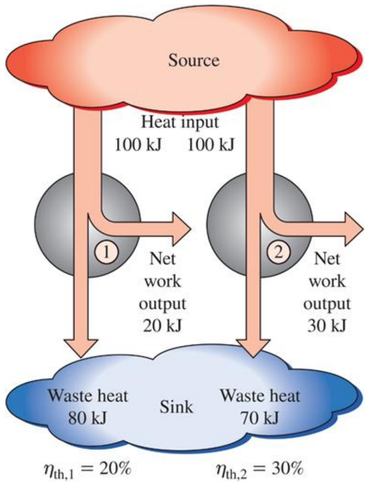


Figure 7–12: Some heat engines perform better than others (convert more of the heat they receive to work).

Heat Engines

Can we save Q_{L} (Q_{out})?

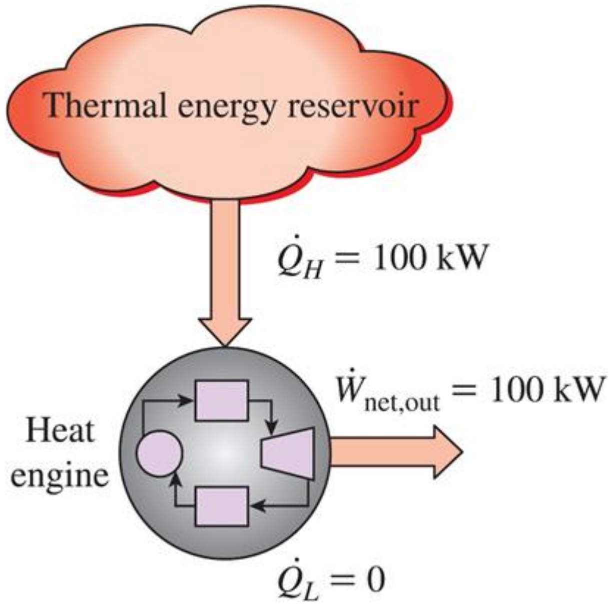
- For a heat engine to operate there **must** be some heat rejected (Q_{L} or Q_{out})
- Often that is a significant amount e.g. if thermal efficiency is 40% then 60% of the heat supplied is rejected as waste heat.
- That waste heat cannot be used usefully in the current cycle however it could be used in other processes. E.g.
  - for another heat engine where it becomes the heat source.
  - Process heat for other industrial operations.

The Kelvin-Planck Statement

- **The Kelvin-Planck statement**:

  It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work

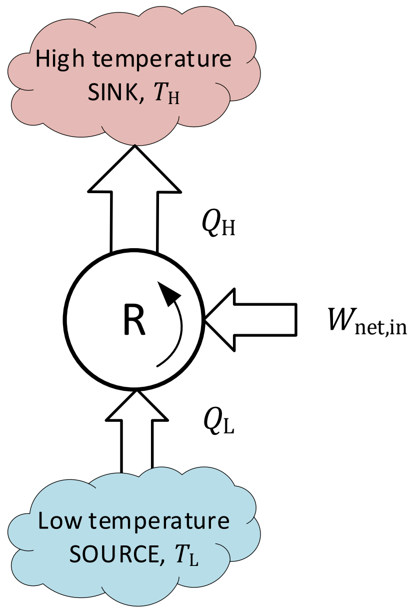
- In other words, it is impossible for Q_{L} to be zero and **impossible for the thermal efficiency to be 100%.**
- Or the thermal efficiency of a heat engine must be less than 100%.
- The Kelvin-Planck statement is a statement of the second law.



This is impossible.
(Okay by the first law)

Refrigerators and Heat Pumps

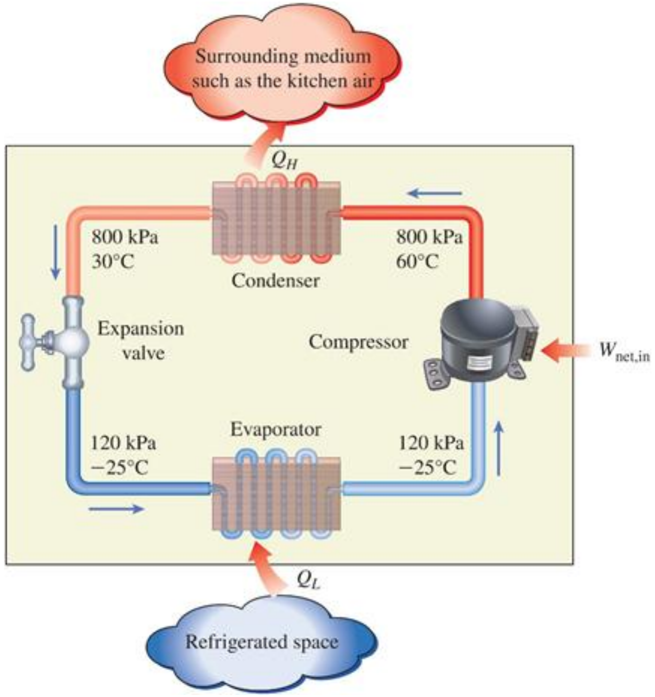
- Heat flows naturally from a high temperature (hot) to a low temperature (cold) but to cause **heat to flow from cold to hot** requires a special device, called a **refrigerator or heat pump.**
- The characteristics of a refrigerator or heat pump are:
    - Absorb heat (Q_{L}) from a low temperature source (at T_{L}).
    - Deliver heat (Q_{H}) to a high temperature sink (at T_{H}).
    - Requires energy input (usually work, $W_{\mathrm{net,in}} = Q_{\mathrm{H}} - Q_{\mathrm{L}}$).
    - Operates in a cycle.
- The working fluid in these cycles is called a **refrigerant**

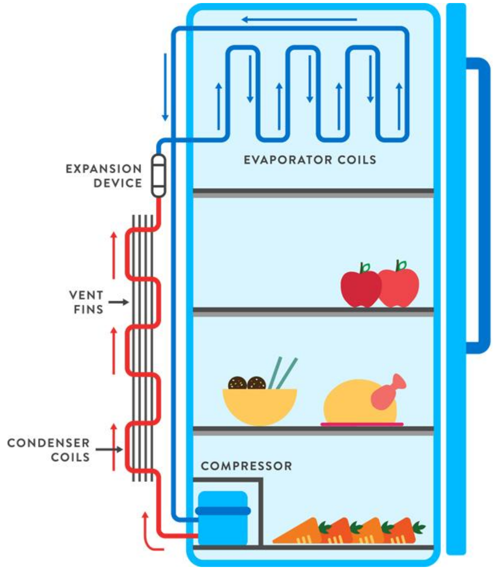


Refrigerators and Heat Pumps

Example: Vapour Compression refrigeration cycle

- Refrigerant enters the compressor as a vapour and is compressed to a higher pressure ($\dot{W}_{\mathrm{net,in}}$).
- It leaves the compressor at relatively high temperature and condenses in the coils of the condenser by rejecting heat to the surroundings ($\dot{Q}_{\mathrm{H}}$).
- It then enters a capillary tube where, as a result of the throttling effect, its pressure and temperature drop dramatically.
- The low temperature liquid refrigerant enters the evaporator where it evaporates by absorbing heat from the refrigerated space ($\dot{Q}_{\mathrm{L}}$).
- The cycle is completed as the refrigerant leaves the evaporator as a vapour and re-enters the compressor.

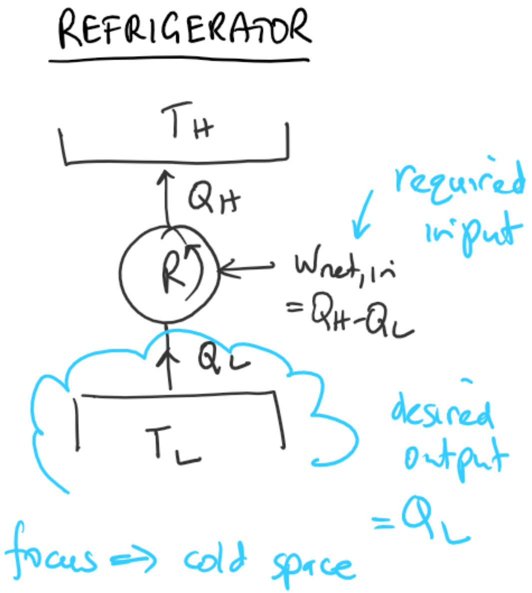


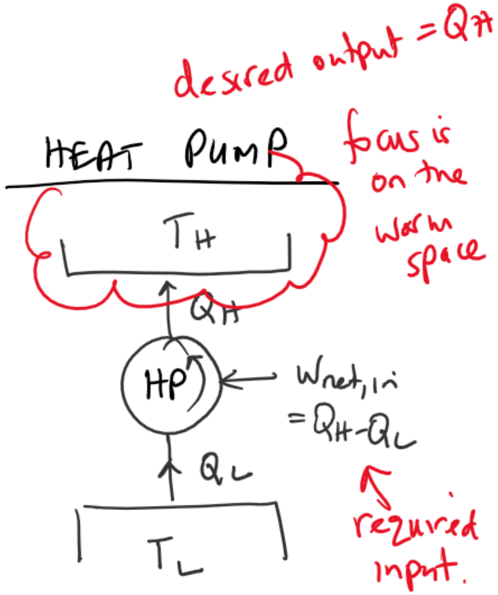


Refrigerators and Heat Pumps

Heat Pump vs Refrigerator, what's the difference?

- They both operate on the same cycle but the difference is in terms of **purpose**:
  - Refrigerator: focus is on the cold space: keeps a cool space cold, so desired output is Q_{L}.
  - Heat pump: focus is on the warm space: keeps a warm space hot, so desired output is Q_{H}.

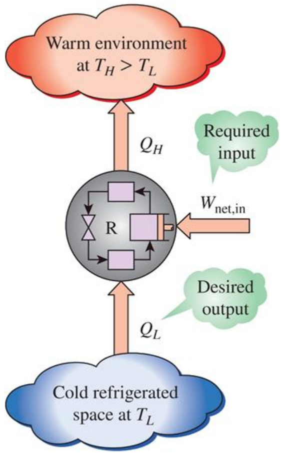




Refrigerators and Heat Pumps

The efficiency of a refrigerator or heat pump is expressed in terms of a coefficient of performance

- **Refrigerator**: objective is to remove heat (Q_L) from the refrigerated space.



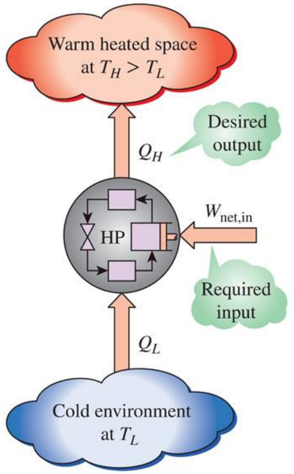
$$COP_\mathrm{R} = \frac{\text{desired output}}{\text{required input}}$$

$$\Rightarrow \quad \boxed{COP_\mathrm{R} = \frac{Q_\mathrm{L}}{W_\mathrm{net,in}}}$$

And since $W_\mathrm{net,out} = Q_\mathrm{H} - Q_\mathrm{L}$

$$\boxed{COP_\mathrm{R} = \frac{Q_\mathrm{L}}{Q_\mathrm{H} - Q_\mathrm{L}} = \frac{1}{\frac{Q_\mathrm{H}}{Q_\mathrm{L}} - 1}}$$

- **Heat pump**: objective is to deliver heat (Q_H) to a warmer space.



$$COP_\mathrm{HP} = \frac{\text{desired output}}{\text{required input}}$$

$$\Rightarrow \quad \boxed{COP_\mathrm{HP} = \frac{Q_\mathrm{H}}{W_\mathrm{net,in}}}$$

And since $W_\mathrm{net,out} = Q_\mathrm{H} - Q_\mathrm{L}$

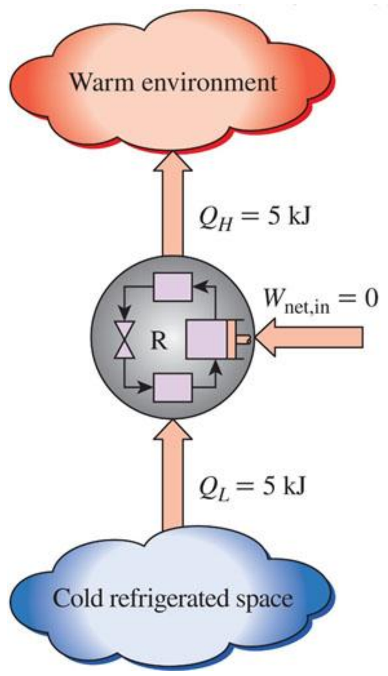
$$\boxed{COP_\mathrm{HP} = \frac{Q_\mathrm{H}}{Q_\mathrm{H} - Q_\mathrm{L}} = \frac{1}{1 - \frac{Q_\mathrm{L}}{Q_\mathrm{H}}}}$$

The Clausius statement

The Clausius Statement:

It is impossible to construct a device that operates in a cycle and produces no effect other than transfer of heat from a lower temperature body to a higher temperature body

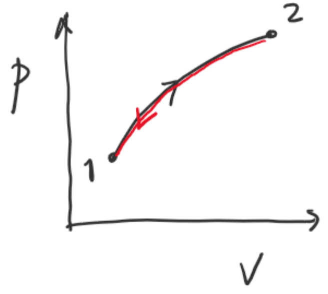
- In other words, it is impossible for a refrigerator or heat pump to operate unless there is some energy input (usually work) to drive the cycle
- It is impossible for the COP to be infinitely large (which it would be if $W_{\mathrm{net,in}}$ was equal to zero: $COP_{\mathrm{R}} = \frac{Q_{\mathrm{L}}}{W_{\mathrm{net,in}}}$).



This is impossible.
(Okay by the first law)

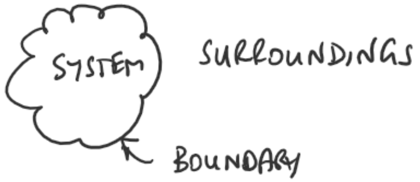
Reversible and Irreversible Processes

- The second law states (Kelvin-Planck statement) that it is impossible for a heat engine to have a thermal efficiency of 100%. So, what is the maximum possible efficiency of a heat engine?
- To answer this question, we need to consider an **idealised process** called a **reversible process**.
- A reversible process is a process that can be **reversed with no net change to the system or surroundings**.
- **All normal, practical processes are <u>irreversible</u>** in that it may be possible to return the system to its original state but there would have been a net change on the surroundings.



1 → 2 a process

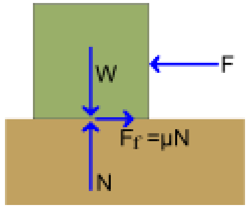
2 → 1 the reverse of that process

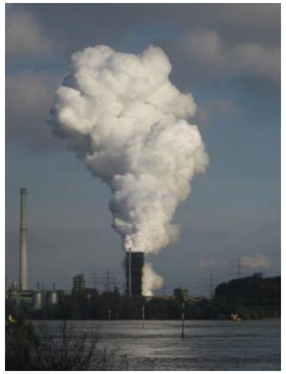


Reversible and Irreversible Processes

Irreversibilities

- These are factors that **cause ALL real processes to be irreversible** and they include:
    - friction
    - unrestrained expansion of gases
    - mixing of two fluids
    - heat transfer across a finite temperature difference
    - combustion, explosions
    - electric resistance
    - chemical reactions etc.
- In real, practical processes one or more of these factors will always be present.

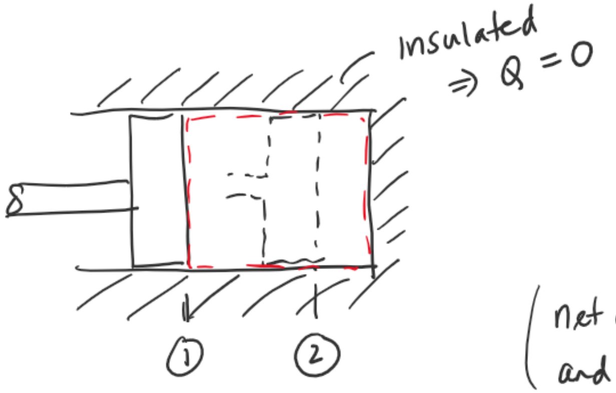




Reversible and Irreversible Processes

Reversible Processes

- The adiabatic compression of a gas in a cylinder comes close to being a reversible process.
- If we assume perfect insulation and frictionless operation, then it is a reversible process.
- In the practical process, friction (and heat loss) leave a net change on the surroundings and so is irreversible.



friction = 1kJ

ideal	(work)	practical	(work)
1 → 2	−8kJ	1 → 2	−9 kJ
2 → 1	+8kJ	2 → 1	+7kJ
net work	0kJ	net change	−2kJ

(net change on system and surroundings = '0')

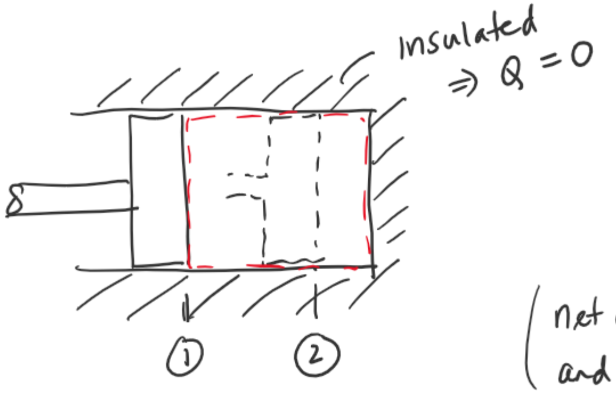
REVERSIBLE PROCESS

(on system and surroundings)

IRREVERSIBLE PROCESS

Reversible and Irreversible Processes

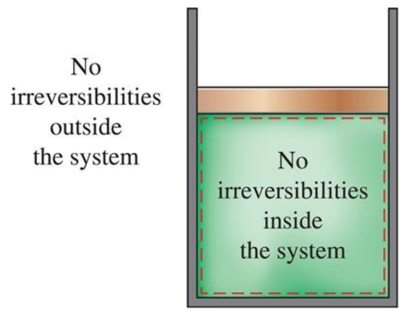
- From this example it can be seen that reversible processes:
  - Consume the least amount of work (when the work is in)
  - Produce the maximum amount of work (when work is out)
- Therefore, a **heat engine cycle based on reversible processes should give us the maximum efficiency possible** (i.e. the theoretical limit for the thermal efficiency of a heat engine).



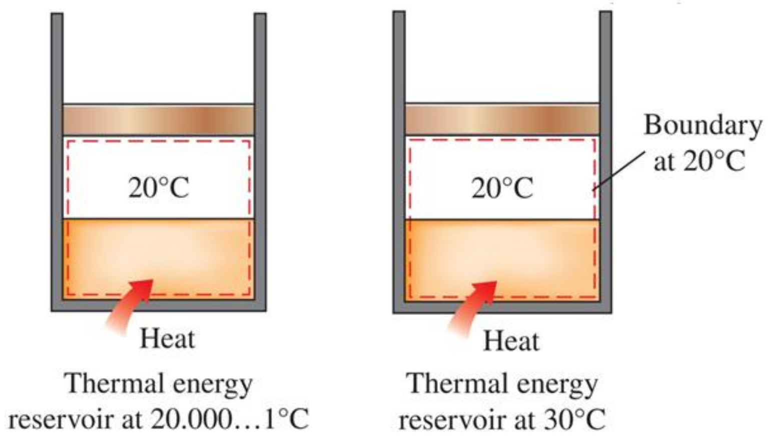
ideal	(work)	practical	(work) friction = 1kJ
1 → 2	− 8kJ	1 → 2	− 9 kJ
2 → 1	+ 8kJ	2 → 1	+ 7kJ
net work	0kJ	net change	− 2kJ
(net change on system and surroundings = '0')		(on system and surroundings)	
REVERSIBLE PROCESS		IRREVERSIBLE PROCESS	

Reversible and Irreversible Processes

- ***Quasi-equilibrium process:*** Defined as a process where the system remains infinitesimally close to the state of equilibrium at all times.
- ***Internally and Externally Reversible Processes***
  - **Internally reversible process:** no irreversibilities occur to the system within the system boundary during the process. (Surroundings may have experienced some irreversibilities)
  - **Externally reversible process**: no irreversibilities occur to the surroundings during the process.
  - **Totally reversible process**: no irreversibilities occur to the system or surroundings during the process
- **For a process to be reversible there must be no irreversibilities and there must be no non-quasi-equilibrium changes**.



A totally reversible process



(*a*) Totally reversible (*b*) Internally reversible

The Carnot Cycle

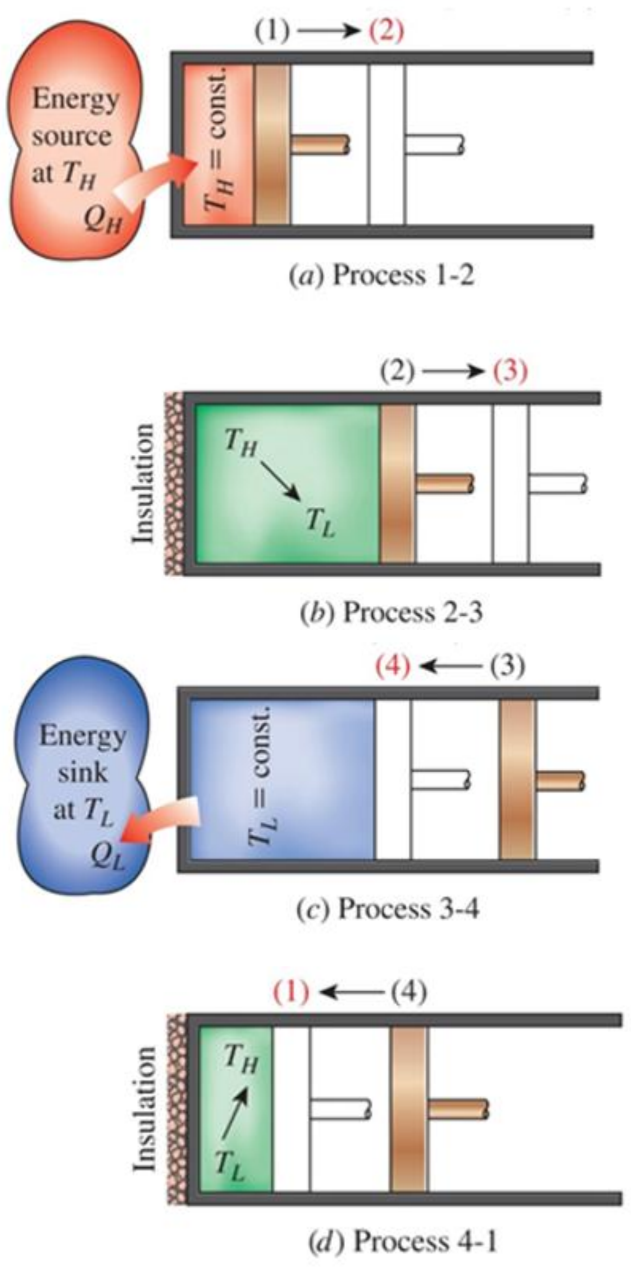
- The Carnot Cycle is a cycle based on reversible processes and is therefore **the most efficient cycle for a heat engine.**
- It is made up of four internally reversible processes:

$1 \rightarrow 2$ **frictionless isothermal absorption of heat** energy at the higher temperature, T_H. Work done, expansion process.

$2 \rightarrow 3$ **frictionless adiabatic expansion** to the lower temperature,T_L. Work done

$3 \rightarrow 4$ **frictionless isothermal rejection of heat** energy at the lower temperature,T_L. Work in, compression process

$4 \rightarrow 1$ **frictionless adiabatic compression** to the higher temperature,T_H.



The Carnot Cycle

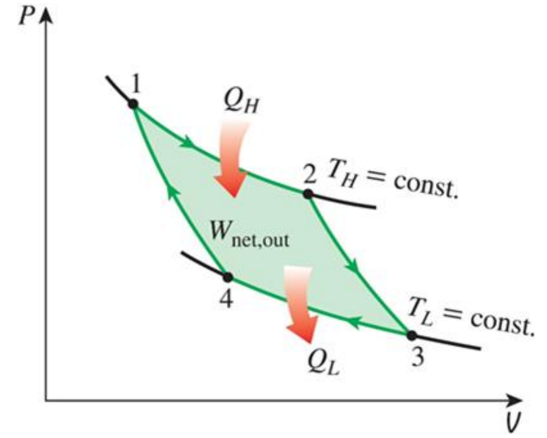
- The Carnot Cycle illustrated on a $P - V$ diagram

$1 \rightarrow 2$ frictionless isothermal absorption of heat energy

$2 \rightarrow 3$ frictionless adiabatic expansion.

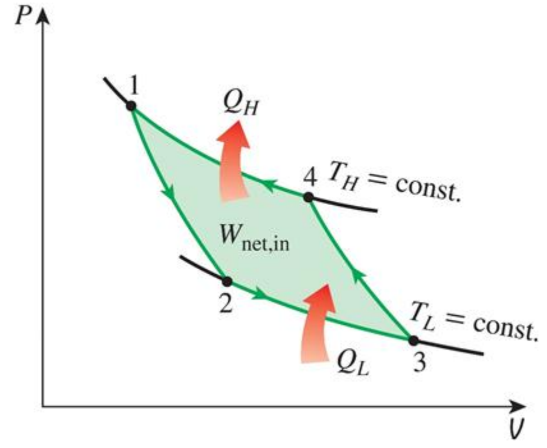
$3 \rightarrow 4$ frictionless isothermal rejection of heat energy

$4 \rightarrow 1$ frictionless adiabatic compression



Carnot Cycle

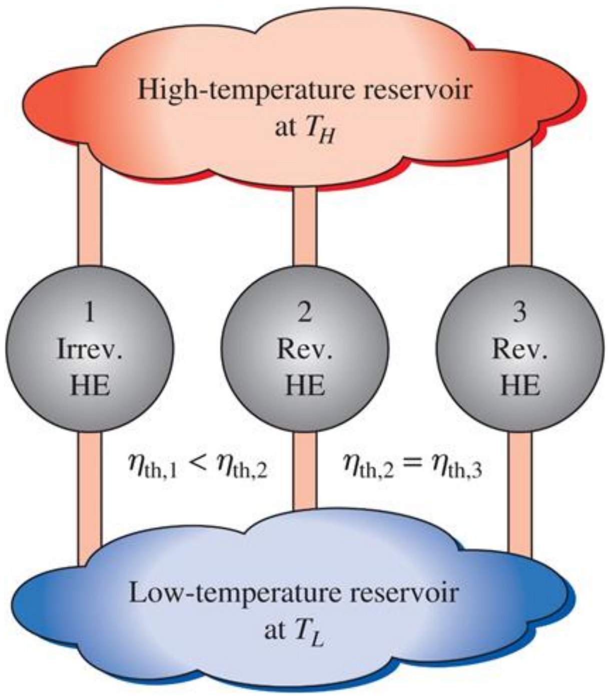
- The Carnot cycle can be reversed (reversible processes!) in which case it becomes The Carnot refrigeration cycle



Reversed Carnot Cycle

The Carnot Principles

- From the Kelvin-Planck and Clausius statements applied to reversible and irreversible heat engines it is possible to express the **Carnot Principles**:
  1. The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.
  2. The efficiencies of all reversible heat engines operating between the same two reservoirs is the same.



Carnot Heat Engine – Carnot Efficiency

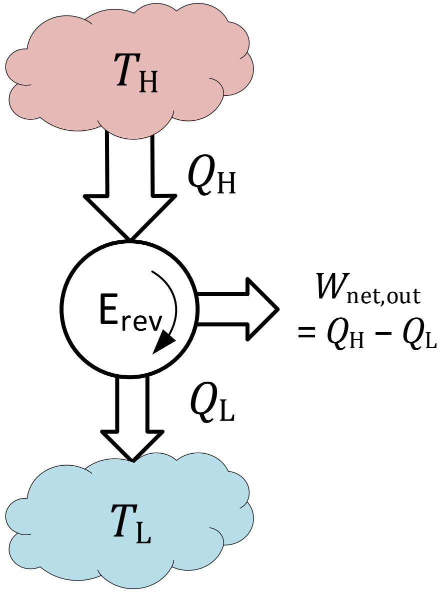
- It is possible to show that for a reversible cycle

$$\left(\frac{\boldsymbol{Q}_{\mathbf{H}}}{\boldsymbol{Q}_{\mathbf{L}}}\right)_{\mathbf{rev}} = \frac{\boldsymbol{T}_{\mathbf{H}}}{\boldsymbol{T}_{\mathbf{L}}}$$

(Where temperatures are in Kelvin)

- Now, the efficiency of *any* heat engine is

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{\text{H}}} = \frac{Q_{\text{H}} - Q_{\text{L}}}{Q_{\text{H}}} = 1 - \frac{Q_{\text{L}}}{Q_{\text{H}}}$$



- So, the **efficiency of a Carnot or reversible heat engine is**:

$$\eta_{\text{th,rev}} = 1 - \left(\frac{Q_{\text{L}}}{Q_{\text{H}}}\right)_{\text{rev}}$$

So

$$\boxed{\eta_{\text{th,rev}} = 1 - \frac{T_{\text{L}}}{T_{\text{H}}}}$$

Carnot Heat Engine – Carnot Efficiency

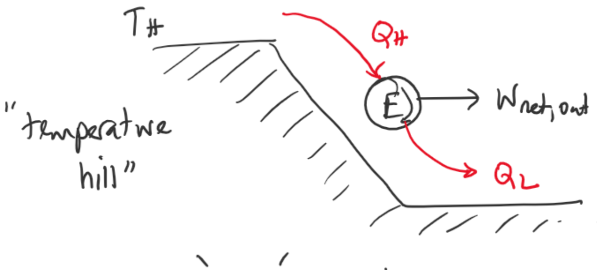
$$\boxed{\eta_{\text{th,rev}} = 1 - \frac{T_{\text{L}}}{T_{\text{H}}}}$$

- This is the highest possible efficiency of a heat engine operating between two thermal energy reservoirs at T_{H} and T_{L}.
- And so, to maximise this ideal efficiency the heat should be absorbed at as high a temperature as possible and should be rejected at as low a temperature as possible. This applies to practical, irreversible heat engines as well.
- Examples
- So, if η_{th} is the thermal efficiency of a heat engine then:

$$\eta_{\text{th}} \begin{cases} < \eta_{\text{th,rev}} & \text{irreversible (i.e. real) heat engine} \\ = \eta_{\text{th,rev}} & \text{reversible heat engine} \\ > \eta_{\text{th,rev}} & \text{impossible heat engine} \end{cases}$$

Quality of Energy

- We see that for the Carnot efficiency, the higher the temperature of the source the greater the thermal efficiency so *as the temperature of the high-temperature source increases more of the high-temperature thermal energy can be converted to useful work*.
- Therefore: **the higher the temperature the higher the quality of the heat energy**.
- And again, **work is a more valuable form of energy transfer than heat** as 100% of work can converted to heat but only a fraction of the heat can be converted to work.



The 'larger' the hill the more efficient the conversion of heat energy into useful work.

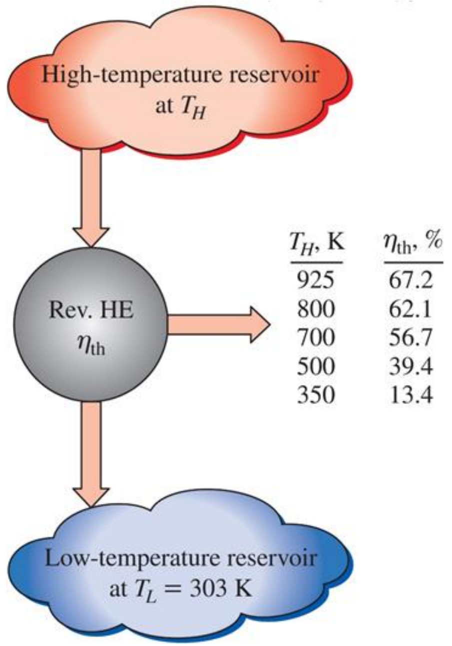


Figure 7–46: The fraction of heat that can be converted to work as a function of source temperature (for T_L = 303 K).

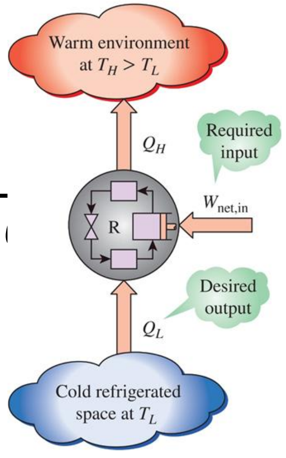
Carnot Refrigerator and Heat Pump

- If we consider a refrigerator or heat pump based on the reversed Carnot cycle, we can determine the maximum possible COP.

- **Refrigerator.**
  For any refrigerator:

$$COP_{\mathrm{R}} = \frac{Q_{\mathrm{L}}}{W_{\mathrm{net,in}}} = \frac{Q_{\mathrm{L}}}{Q_{\mathrm{H}} - }$$

$$COP_{\mathrm{R}} = \frac{1}{\frac{Q_{\mathrm{H}}}{Q_{\mathrm{L}}} - 1}$$



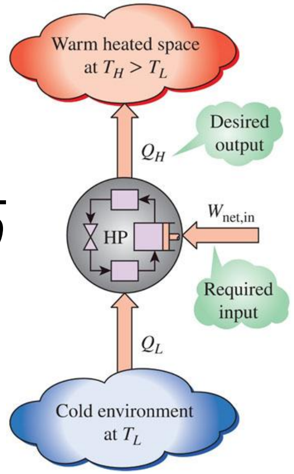
  So for a **reversible refrigerator**

$$\boxed{COP_{\mathrm{R,rev}} = \frac{1}{\frac{T_{\mathrm{H}}}{T_{\mathrm{L}}} - 1}}$$

- **Heat pump**.
  For any heat pump:

$$COP_{\mathrm{HP}} = \frac{Q_{\mathrm{H}}}{W_{\mathrm{net,in}}} = \frac{Q_{\mathrm{H}}}{Q_{\mathrm{H}} - Q}$$

$$COP_{\mathrm{HP}} = \frac{1}{1 - \frac{Q_{\mathrm{L}}}{Q_{\mathrm{H}}}}$$



  So for a **reversible heat pump**

$$\boxed{COP_{\mathrm{HP,rev}} = \frac{1}{1 - \frac{T_{\mathrm{L}}}{T_{\mathrm{H}}}}}$$

Carnot Refrigerator and Heat Pump

$$COP_{\mathrm{R,rev}} = \frac{1}{\frac{T_\mathrm{H}}{T_\mathrm{L}} - 1} \qquad\qquad COP_{\mathrm{HP,rev}} = \frac{1}{1 - \frac{T_\mathrm{L}}{T_\mathrm{H}}}$$

- These COP's represent the highest possible for a refrigerator or heat pump operating between the temperature limits of at T_L and T_H
- And so, to maximise these ideal coefficients of performance the difference between T_L and T_H should be as low as possible (a small "temperature hill"). In other words, the COP will reduce as the temperature difference increases. This applies to practical, irreversible heat engines as well.

Carnot Refrigerator and Heat Pump

- So, if COP_{R} is the coefficient of performance for a refrigerator then:

$$COP_{\mathrm{R}} \begin{cases} < COP_{\mathrm{R,rev}} & \text{irreversible (i.e. real) refrigerator} \\ = COP_{\mathrm{R,rev}} & \text{reversible refrigerator} \\ > COP_{\mathrm{R,rev}} & \text{impossible refrigerator} \end{cases}$$

- Similarly for a heat pump:

$$COP_{\mathrm{HP}} \begin{cases} < COP_{\mathrm{HP,rev}} & \text{irreversible (i.e. real) heat pump} \\ = COP_{\mathrm{HP,rev}} & \text{reversible heat pump} \\ > COP_{\mathrm{HP,rev}} & \text{impossible heat pump} \end{cases}$$